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CURRENT INTRODUCTION LINE AND SUPERCONDUCTING MAGNET DEVICE

Abstract

A current introduction line for introducing a current into a superconducting coil in a vacuum vessel, includes: a vacuum feedthrough; a busbar that is disposed in the vacuum vessel and is electrically connected to the superconducting coil; and a rigid conductor that is fixed to the vacuum vessel so as to be thermally coupled to the vacuum vessel and to be electrically insulated from the vacuum vessel, and electrically connects the vacuum feedthrough to the busbar.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This is a bypass continuation of International PCT Application No. PCT/JP2023/036219, filed on Oct. 4, 2023, which claims priority to Japanese Patent Application No. 2022-204676, filed on Dec. 21, 2022, which are incorporated by reference herein in their entirety.

BACKGROUND
Technical Field
[0002] Certain embodiments of the present invention relate to a current introduction line and a superconducting magnet device.
Description of Related Art
[0003] In general, a superconducting magnet device includes a superconducting coil and a vacuum vessel that accommodates the superconducting coil in a state of being cooled to a cryogenic temperature. In order to supply power to the superconducting coil from the outside, a coil electrode is provided outside the vacuum vessel. The coil electrode is cooled by thermal conduction from the superconducting coil. Therefore, moisture in the air surrounding the vacuum vessel may adhere to or freeze on the coil electrode. In the related art, in order to prevent such condensation, it is known to blow heated air onto the coil electrode to heat the coil electrode.
SUMMARY
[0004] According to an embodiment of the present invention, there is provided a current introduction line for introducing a current into a superconducting coil in a vacuum vessel. The current introduction line includes: a vacuum feedthrough; a busbar that is disposed in the vacuum vessel and is electrically connected to the superconducting coil; and a rigid conductor that is fixed to the vacuum vessel so as to be thermally coupled to the vacuum vessel and to be electrically insulated from the vacuum vessel, and electrically connects the vacuum feedthrough to the busbar.
[0005] According to another embodiment of the present invention, there is provided a superconducting magnet device including a vacuum vessel, a superconducting coil disposed in the vacuum vessel, and a current introduction line for introducing a current into the superconducting coil. The current introduction line includes a vacuum feedthrough, a busbar that is disposed in the vacuum vessel and is electrically connected to the superconducting coil, and a rigid conductor that is fixed to the vacuum vessel so as to be thermally coupled to the vacuum vessel and to be electrically insulated from the vacuum vessel, and electrically connects the vacuum feedthrough to the busbar.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0006] FIG. **1** is a diagram schematically showing a superconducting magnet device according to an embodiment.
[0007] FIG. **2** is a perspective view schematically showing a current introduction line according to the embodiment.
[0008] FIG. **3** is a side view schematically showing another example of the current introduction

line according to the embodiment.

DETAILED DESCRIPTION

[0009] It is desirable to suppress condensation on a current introduction line of a superconducting magnet device with a simple configuration.

[0010] Hereinafter, an embodiment for carrying out the present invention will be described in detail with reference to the drawings. In the description and drawings, identical or equivalent components, members, and processing are denoted by the same reference numerals, and overlapping description is omitted as appropriate. The scale or shape of each part that is shown in the drawings is conveniently set for ease of description and is not limitedly interpreted unless otherwise specified. The embodiment is merely an example and does not limit the scope of the present invention in any way. All characteristics and combinations to be described in the embodiment are not necessarily essential to the invention.

[0011] FIG. **1** is a diagram schematically showing a superconducting magnet device **10** according to an embodiment. The superconducting magnet device **10** can be mounted on a high-magnetic field utilization device as a magnetic field source of, for example, a single crystal pulling device, a nuclear magnetic resonance (NMR) system, a magnetic resonance imaging (MRI) system, an accelerator such as a cyclotron, a high energy physics system such as a nuclear fusion system, or other high-magnetic field utilization devices (not shown) and can generate a high magnetic field required for the device.

[0012] The superconducting magnet device **10** includes a superconducting coil **12**, a vacuum vessel **14**, a heat shield **16**, and a current introduction line **20**.

[0013] The superconducting coil **12** is disposed in the vacuum vessel **14**, and is configured to generate a strong magnetic field by being energized in a state of being cooled to a cryogenic temperature equal to or lower than a superconducting transition temperature. The superconducting coil **12** may be a known superconducting coil (for example, a so-called low-temperature superconducting coil).

[0014] The superconducting coil **12** is connected to an external power source **18** disposed outside the vacuum vessel **14** by a current introduction line **20**. An excitation current is supplied from the external power source **18** to the superconducting coil **12** through the current introduction line **20**.

[0015] The superconducting coil **12** is thermally coupled to a cryocooler **15**, for example, a two-stage Gifford-McMahon (GM) cryocooler or other types of cryocoolers installed in the vacuum vessel **14**, and is used in a state of being cooled to a cryogenic temperature equal to or lower than a superconducting transition temperature. In this embodiment, the superconducting magnet device **10** is configured as a so-called conduction-cooled type in which the superconducting coil **12** is directly cooled by the cryocooler **15**, instead of an immersion-cooled type in which the superconducting coil **12** is immersed in a cryogenic liquid refrigerant such as liquid helium. The superconducting magnet device **10** may be of an immersion-cooled type.

[0016] The vacuum vessel **14** is an adiabatic vacuum vessel that provides a cryogenic vacuum environment suitable for bringing the superconducting coil **12** into a superconducting state, and is also called a cryostat. Typically, the vacuum vessel **14** has a columnar shape or a cylindrical shape with a hollow portion at a central portion thereof. Therefore, the vacuum vessel **14** includes a substantially flat circular or annular top plate **14***a* and bottom plate **14***b*, and a cylindrical side wall (cylindrical outer peripheral wall, or coaxially disposed cylindrical outer peripheral wall and inner peripheral wall) connecting the top plate **14***a* and the bottom plate **14***b*. The cryocooler **15** may be installed on the top plate **14***a* of the vacuum vessel **14**. The vacuum vessel **14** is formed of, for example, a metal material such as stainless steel or other suitable high-strength materials to withstand an ambient pressure (for example, atmospheric pressure).

[0017] The heat shield **16** is disposed to surround the superconducting coil **12** within the vacuum vessel **14**. The heat shield **16** is formed of, for example, a metal material such as copper or other materials having high thermal conductivity. The heat shield **16** may be cooled by a first-stage cooling stage of the two-stage cryocooler **15** that cools the superconducting coil **12**, or by a single-stage cryocooler separate from the two-stage cryocooler. During an operation of the superconducting magnet device **10**, the heat shield **16** is cooled to a first cooling temperature, for example, 30 K to 50 K, and the superconducting coil **12** is cooled to a second cooling temperature lower than the first cooling temperature, for example, 3 K to 20 K (for example, about 4 K). The heat shield **16** can thermally protect a low-temperature section such as the superconducting coil **12**, which is disposed inside the heat shield **16** and is cooled to a lower temperature than the heat shield **16**, from radiant heat from the vacuum vessel **14**.

[0018] The current introduction line **20** for introducing a current into the superconducting coil **12** includes an external wire **22**, a vacuum feedthrough **24**, a rigid conductor **26**, a busbar **28**, and a current lead portion **30**, and forms a current path from the external power source **18** to the superconducting coil **12**. For simplicity, only one current introduction line **20** is shown in FIG. **1**. However, in general, a plurality of the current introduction lines **20** may be provided in the superconducting magnet device **10**, and for example, one current introduction line **20** on a positive electrode side and one current introduction line **20** on a negative electrode side may be provided.

[0019] The external wire **22** disposed outside the vacuum vessel **14** connects the external power source **18** to the vacuum feedthrough **24** provided in a wall portion of the vacuum vessel **14**. The external wire **22** may be a suitable power supply cable.

[0020] The vacuum feedthrough **24** is a hermetically sealed terminal for introducing a current into the vacuum vessel **14**, and connects the external wire **22** to internal wires (that is, the rigid conductor **26**, the busbar **28**, and the current lead portion **30**) inside the vacuum vessel **14**. The current introduction line **20** can penetrate the wall portion of the vacuum vessel **14** while maintaining airtightness of the vacuum vessel **14** using the vacuum feedthrough **24**. The vacuum feedthrough **24** is exposed to an ambient environment (for example, room temperature and atmospheric pressure environment) of the vacuum vessel **14**, and thus is at an ambient temperature (for example, room temperature).

[0021] In the embodiment, as shown, the vacuum feedthrough **24** is installed on the bottom plate **14***b* of the vacuum vessel **14**, and the current introduction line **20** is disposed at an outer peripheral portion of the vacuum vessel **14** on a lower side. This disposition is advantageous from the viewpoint of workability. Depending on the application of the superconducting magnet device **10**, the vacuum vessel **14** is often considerably larger than a worker (for example, several meters or more in diameter). When the vacuum feedthrough **24** is provided at the outer peripheral portion of the vacuum vessel **14**, it is easy for the worker to access the current introduction line **20** from around the superconducting magnet device **10**. The vacuum feedthrough **24** may be installed on an upper surface of the vacuum vessel **14**, and the current introduction line **20** may be disposed at the outer peripheral portion of the vacuum vessel **14** on an upper side. Alternatively, the vacuum feedthrough **24** and the current introduction line **20** may be provided at other portions of the vacuum vessel **14**.

[0022] The rigid conductor **26** and the busbar **28** are disposed outside the heat shield **16** in the vacuum vessel **14**, and connect the vacuum feedthrough **24** to the current lead portion **30**. The vacuum feedthrough **24**, the rigid conductor **26**, and the busbar **28** are formed of a conductive material, for example, a metal material having excellent conductivity typified by pure copper such as oxygen-free copper, and serve as the current path to the superconducting coil **12**.

[0023] Although details will be described later, the rigid conductor **26** is fixed to the vacuum vessel **14** so as to be thermally coupled to the vacuum vessel **14** and to be electrically insulated from the vacuum vessel **14**, and electrically connects the vacuum feedthrough **24** to the busbar **28**. The busbar **28** is electrically connected to the superconducting coil **12** through the current lead portion **30**.

[0024] The busbar **28** may be thermally coupled to the heat shield **16** in a state of being electrically insulated from the heat shield **16**. An end portion of the busbar **28** on a side opposite to the rigid conductor **26** may be fixed to the heat shield **16** or may be connected to the heat shield **16** via an appropriate heat transfer member to be cooled to the first cooling temperature in the same manner as the heat shield **16**. Therefore, the current introduction line **20** can have a temperature distribution in which the temperature gradually decreases from the vacuum feedthrough **24** to the rigid conductor **26** and the busbar **28**.

[0025] The rigid conductor **26** and the busbar **28** may be disposed outside a thermal insulation layer **17**. The thermal insulation layer **17** may be provided between the vacuum vessel **14** and the heat shield **16** in order to protect the heat shield **16** from radiant heat emitted from the vacuum vessel **14**. The thermal insulation layer **17** may be, for example, a multilayer insulation (MLI), and may be disposed to surround the heat shield **16**.

[0026] In this embodiment, the busbar **28** is a rigid conductor separate from the rigid conductor **26**. However, in a certain embodiment, the rigid conductor **26** and the busbar **28** may be configured as a single conductor member.

[0027] The current lead portion **30** is disposed inside the heat shield **16**, and connects the busbar **28** to the superconducting coil **12**. The current lead portion **30** may extend in a different direction within the vacuum vessel **14** than the busbar **28**. In the shown example, the current lead portion **30** may extend from the end portion of the busbar **28** to the superconducting coil **12** in a longitudinal direction (vertical direction). The current lead portion **30** may include terminal portions at both ends respectively connected to the busbar **28** and the superconducting coil **12**, and a superconducting current lead connecting the terminal portions. The superconducting current lead may have, for example, a rod shape such as a columnar shape, and may be formed of a copper oxide superconductor or other high-temperature superconducting materials. Alternatively, the superconducting current lead may be formed of a low-temperature superconducting material represented by NbTi.

[0028] FIG. **2** is a perspective view schematically showing the current introduction line **20** according to the embodiment. As described above, the current introduction line **20** includes the vacuum feedthrough **24**, the rigid conductor **26**, and the busbar **28** that are electrically connected to each other.

[0029] The vacuum feedthrough **24** includes a feedthrough conductor **24***a* and a feedthrough insulating material **24***b*. The feedthrough conductor **24***a* is connected to the external wire **22** at one end exposed to the outside of the vacuum vessel **14**, and is connected to the rigid conductor **26** at the other end inside the vacuum vessel **14**. In this way, a current path from the external wire **22** to the rigid conductor **26** through the feedthrough conductor **24***a* is formed.

[0030] The feedthrough conductor **24***a* is inserted through an opening portion of the vacuum vessel **14** (the bottom plate **14***b* in this example). The opening portion is filled with the feedthrough insulating material **24***b* to maintain the airtightness of the vacuum vessel **14**, and the feedthrough conductor **24***a* is installed in the vacuum vessel **14** in a state of being electrically insulated from the vacuum vessel **14** by the feedthrough insulating material **24***b*. The feedthrough insulating material **24***b* may be, for example, any suitable electrically insulating material such as a synthetic resin material.

[0031] The rigid conductor **26** includes a first end portion **26***a*, a conductor plate **26***b*, a deformable portion **26***c*, and a second end portion **26***d*.

[0032] In this embodiment, the rigid conductor **26** is formed of a single plate made of a conductive material (for example, copper) that is bent to form the first end portion **26***a*, the conductor plate **26***b*, the deformable portion **26***c*, and the second end portion **26***d*. The conductor plate **26***b* is a flat plate-shaped portion fixed to a fixing surface of the vacuum vessel **14** as will be described later, and has, for example, a rectangular outer shape. The conductor plate **26***b* may have an outer shape of another shape.

[0033] The first end portion **26***a* extends from a part of an outer periphery of the conductor plate **26***b*, is bent so as to rise with respect to the fixing surface of the vacuum vessel **14**, and is further bent from a rising portion in a direction along the fixing surface of the vacuum vessel **14** toward the feedthrough conductor **24***a*. Similarly, the second end portion **26***d* extends from a part of the outer periphery of the conductor plate **26***b* on a side opposite to the first end portion **26***a*, is bent to rise with respect to the fixing surface of the vacuum vessel **14**, and is further bent from a rising portion in the direction along the fixing surface of the vacuum vessel **14** toward the busbar **28**. The rising portion of the second end portion **26***d* serves as the deformable portion **26***c* as will be described later.

[0034] The first end portion **26***a*, the conductor plate **26***b*, the deformable portion **26***c*, and the second end portion **26***d* are prepared as individual pieces, and are joined together by an appropriate joining method such as fastening, soldering, or brazing, and may be integrated into the rigid conductor **26**.

[0035] The rigid conductor **26** is fixed to the vacuum feedthrough **24** at the first end portion **26***a*, and is fixed to the busbar **28** at the second end portion **26***d*. The first end portion **26***a* is fixed to the feedthrough conductor **24***a* by a fastening member **32** such as a bolt, and the conductor plate **26***b* is fixed to the busbar **28** by the fastening member **32**. The first end portion **26***a* and the second end portion **26***d* may be respectively joined to the vacuum feedthrough **24** and the busbar **28** by an appropriate joining method such as soldering or brazing. In this way, a current path from the vacuum feedthrough **24** to the busbar **28** through the rigid conductor **26** is formed.

[0036] The conductor plate **26***b* is fixed to the vacuum vessel **14** so as to sandwich an electrical insulation layer **34** between the vacuum vessel **14** and the conductor plate **26***b*. The electrical insulation layer **34** may be, for example, a sheet or a plate formed of a synthetic resin material such as glass fiber reinforced plastic (GFRP) or other suitable electrical insulating materials. One surface of the electrical insulation layer **34** is in contact with the fixing surface of the vacuum vessel **14** and the opposite surface of the electrical insulation layer **34** is in contact with a main surface of the conductor plate **26***b* such that the electrical insulation layer **34** is sandwiched between the vacuum vessel **14** and the conductor plate **26***b*. Therefore, the conductor plate **26***b*, that is, the rigid conductor **26** is not in contact with the vacuum vessel **14**, and is in a state of being electrically insulated from the vacuum vessel **14**. As shown, the conductor plate **26***b* may be fixed to the vacuum vessel **14** by the fastening member **32** such as a bolt. In this case, in order to prevent electrical conduction between the conductor plate **26***b* and the vacuum vessel **14** through the fastening member **32** and to ensure electrical insulation between the conductor plate **26***b* and the vacuum vessel **14**, the fastening member **32** may be formed of, for example, a synthetic resin material such as an engineering plastic or other suitable electrically insulating materials. In addition, in a case where a metal fastening member **32** is used, a washer made of an electrically insulating material may be interposed between a head portion of the fastening member **32** and the conductor plate **26***b*. In this way, the conductor plate **26***b*, that is, the rigid conductor **26** is fixed to the vacuum vessel **14** in a state of being electrically insulated from the vacuum vessel **14**.

[0037] Alternatively, the conductor plate **26***b* may be joined to the electrical insulation layer **34** by any suitable joining method such as bonding. Similarly, the electrical insulation layer **34** may be joined to the vacuum vessel **14** by any suitable joining method.

[0038] The conductor plate **26***b* is thermally coupled to the vacuum vessel **14** via the electrical insulation layer **34**. Therefore, heat can flow from the vacuum vessel **14** to the conductor plate **26***b* through the electrical insulation layer **34**. For better thermal connection, it is desirable that a thickness of the electrical insulation layer **34** is as thin as possible. However, in order to ensure the electrical insulation between the conductor plate **26***b* and the vacuum vessel **14**, it is preferable that the electrical insulation layer **34** is relatively thick. Therefore, the thickness of the electrical insulation layer **34** may be selected from, for example, a range of 1 mm to 10 mm. In a case where the electrical insulation layer **34** is a thick plate, input heat from the vacuum vessel **14** to the conductor plate **26***b* can also be increased by increasing an area of the electrical insulation layer **34** (and the conductor plate **26***b*) that is in contact with the fixing surface of the vacuum vessel **14**.

[0039] While the rigid conductor **26** is directly fixed to the vacuum vessel **14** as described above, the busbar **28** is disposed at an interval from the fixing surface of the vacuum vessel **14** to which the rigid conductor **26** is fixed, and is fixed to the vacuum vessel **14** via the rigid conductor **26**. The busbar **28** is not in contact with the vacuum vessel **14**.

[0040] As shown, the busbar **28** may extend along the fixing surface (for example, the bottom plate **14***b*) of the vacuum vessel **14**, or may extend in a lateral direction (horizontal direction) within the vacuum vessel **14** in a case where the vacuum vessel **14** is disposed such that the top plate **14***a* faces upward and the bottom plate **14***b* faces downward. In addition, depending on the disposition of internal devices such as the superconducting coil **12** and the heat shield **16** in the vacuum vessel **14**, the busbar **28** may extend in another direction, for example, a longitudinal direction (the vertical direction) in the vacuum vessel **14**. The busbar **28** may be, for example, a thin plate or a bar having a belt-like or rectangular cross-sectional shape.

[0041] As described above, the busbar **28** can be cooled by thermal conduction from the low-temperature section, such as the heat shield **16**. Contrary to this, although the conductor plate **26***b* is connected to the busbar **28**, the conductor plate **26***b* is not cooled as much as the busbar **28** due to input heat from the surrounding environment caused by thermal coupling with the vacuum vessel **14**. Since the vacuum vessel **14** has an ambient temperature (for example, room temperature), the conductor plate **26***b* can be maintained at the ambient temperature or a temperature close to the ambient temperature.

[0042] The vacuum feedthrough **24** is exposed to the surrounding environment, and thus, is at the ambient temperature, similar to the vacuum vessel **14**. The thermal coupling between the conductor plate **26***b* and the vacuum vessel **14** can keep the temperature of the conductor plate **26***b* at the same level as the temperature of the vacuum feedthrough **24**, and can reduce a temperature difference between the conductor plate **26** and the vacuum vessel **14** to be smaller than a temperature difference between the low-temperature section such as the heat shield **16** and the rigid conductor **26**. The rigid conductor **26** can reduce an influence of the temperature decrease of the busbar **28** due to the thermal conduction from the low-temperature section on the vacuum feedthrough **24**.

[0043] Therefore, according to the embodiment, by providing the rigid conductor **26** thermally coupled to the vacuum vessel **14** in the current introduction line **20**, it is possible to suppress the temperature decrease of the vacuum feedthrough **24** and prevent or reduce the condensation from the surrounding environment onto the vacuum feedthrough **24**. Compared to existing countermeasures for condensation using the installation of a heater or the blowing of heated air, it is possible to suppress condensation on the current introduction line **20** of the superconducting magnet device **10** with a simple configuration.

[0044] In addition, the rigid conductor **26** is provided with the deformable portion **26***c* attached to the busbar **28** so that the conductor plate **26***b* is electrically connected to the busbar **28**. As described above, the deformable portion **26***c* is bent to rise from the conductor plate **26***b* with respect to the fixing surface of the vacuum vessel **14**. Therefore, when the busbar **28** and the second end portion **26***d* are displaced in an extending direction of the busbar **28** (for example, a left-right direction in the figure), the deformable portion **26***c* can be bent with respect to the conductor plate **26***b*.

[0045] The low-temperature section such as the heat shield **16** and the busbar **28** can thermally contract by cooling. Accordingly, the busbar **28** can be pulled toward the low-temperature section as schematically shown by an arrow **36** in FIG. **2**. In this case, the deformable portion **26***c* is capable of being bent in response to such deformation of the busbar **28**, as schematically shown by an arrow **38**. The displacement of the busbar **28** due to thermal contraction is absorbed by the deformable portion **26***c*. Since the rigid conductor **26** is fixed to the vacuum vessel **14** by the conductor plate **26***b*, the vacuum feedthrough **24** is less susceptible to an influence of such thermal contraction. It is possible to prevent the occurrence of an excessive thermal stress in, for example, the feedthrough insulating material **24***b* of the vacuum feedthrough **24**, and ultimately the occurrence of unexpected deformation or damage in the vacuum feedthrough **24** and/or the vacuum vessel **14**.

[0046] The deformable portion **26***c* is not limited to a bent portion. The deformable portion **26***c* may have other structures formed in the rigid conductor **26** and deformable in response to the displacement of the busbar **28**.

[0047] The present invention has been described above based on the embodiment. It will be understood by those skilled in the art that the present invention is not limited to the above embodiments, various design changes can be made, various modification examples are possible, and such modification examples are also within the scope of the present invention. Various features described in relation to a certain embodiment are also applicable to other embodiments. A new embodiment generated through combination also has the effects of each of the combined embodiments.

[0048] In the above-described embodiment, a case where the rigid conductor **26** is directly fixed to a wall surface of the vacuum vessel **14** has been described as an example. However, the rigid conductor **26** may be fixed to another part in the vacuum vessel **14** as exemplified below.

[0049] FIG. **3** is a side view schematically showing another example of the current introduction line **20** according to the embodiment. As in the above-described embodiment, the current introduction line **20** also includes the vacuum feedthrough **24**, the rigid conductor **26**, and the busbar **28** in the embodiment of FIG. **3**. The rigid conductor **26** electrically connects the vacuum feedthrough **24** to the busbar **28**.

[0050] However, the rigid conductor **26** is fixed to a support member **40** so as to sandwich the electrical insulation layer **34** between the support member **40** and the rigid conductor **26**, and is thermally coupled to the support member **40** via the electrical insulation layer **34**. The support member **40** is formed of a metal material such as stainless steel, and may have a plate shape or other shapes. For example, the support member **40** may be fixed to a pedestal **42** fixed to the wall surface of the vacuum vessel **14** in the vacuum vessel **14**. A support body **44** that supports the superconducting coil **12** may be attached to the pedestal **42**. In this way, the rigid conductor **26** may be thermally coupled to the vacuum vessel **14** via the support member **40**.

[0051] In addition, in the above-described embodiment, a case where the rigid conductor **26** is a plate member made of a conductive material has been described as an example. However, the rigid conductor **26** is not limited to such a plate, and may be, for example, a rod or a block of conductor.

[0052] Although the present invention has been described using specific terms based on the embodiment, the embodiment only shows one aspect of the principle and application of the invention, and the embodiment allows for many modifications and changes in arrangement without departing from the concept of the invention as defined in the claims.

[0053] The present invention can be used in the field of a current introduction line and a superconducting magnet device.

[0054] It should be understood that the invention is not limited to the above-described embodiment, but may be modified into various forms on the basis of the spirit of the invention. Additionally, the modifications are included in the scope of the invention.

Claims

1. A current introduction line for introducing a current into a superconducting coil in a vacuum vessel, the current introduction line comprising: a vacuum feedthrough; a busbar that is disposed in the vacuum vessel and is electrically connected to the superconducting coil; and a rigid conductor that is fixed to the vacuum vessel so as to be thermally coupled to the vacuum vessel and to be electrically insulated from the vacuum vessel, and electrically connects the vacuum feedthrough to the busbar.

2. The current introduction line according to claim 1, wherein the vacuum feedthrough is a hermetically sealed terminal for introducing a current into the vacuum vessel, and connects an external wire to an internal wire inside the vacuum vessel.
3. The current introduction line according to claim 2, wherein the vacuum feedthrough includes a feedthrough conductor and a feedthrough insulating material, and the feedthrough conductor is connected to the external wire at one end exposed to an outside of the vacuum vessel, and is connected to the rigid conductor at the other end inside the vacuum vessel.
4. The current introduction line according to claim 3, wherein the feedthrough conductor is installed in the vacuum vessel in a state of being electrically insulated from the vacuum vessel by the feedthrough insulating material made of an electrically insulating material.
5. The current introduction line according to claim 1, wherein the rigid conductor includes a conductor plate, and the conductor plate is fixed to the vacuum vessel so as to sandwich an electrical insulation layer between the vacuum vessel and the conductor plate, and is thermally coupled to the vacuum vessel via the electrical insulation layer.
6. The current introduction line according to claim 1, wherein the rigid conductor includes a deformable portion that is attached to the busbar so as to electrically connect the rigid conductor to the busbar.
7. The current introduction line according to claim 6, wherein the deformable portion is bent to rise from a conductor plate of the rigid conductor with respect to a fixing surface of the vacuum vessel.
8. A superconducting magnet device comprising: a vacuum vessel; a superconducting coil disposed in the vacuum vessel; and a current introduction line for introducing a current into the superconducting coil, wherein the current introduction line includes a vacuum feedthrough, a busbar that is disposed in the vacuum vessel and is electrically connected to the superconducting coil, and a rigid conductor that is fixed to the vacuum vessel so as to be thermally coupled to the vacuum vessel and to be electrically insulated from the vacuum vessel, and electrically connects the vacuum feedthrough to the busbar.